INTELLI-CREDIT: ARCHITECTURE DEEP DIVE

Technical breakdown of all AI models, extraction engines, and processing pipelines

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IntelliCredit is not a single wrapper around an LLM. It is a multi-agent orchestrated pipeline containing 8 specialized engines. This document details the technical implementation, algorithmic choices, and the specific models powering each engine.

1. Data Ingestion & Preprocessing Engine

Role: Extracts raw text and tabular structures from unstructured PDFs, scanned images, and Excel/CSV files (GST & Bank statements).

Technical Implementation & Models Used:

- **Primary PDF Extraction:** *PyMuPDF (fitz)*. Specifically chosen for its speed and spatial preservation of text blocks in native PDFs.
- **Tabular Extraction:** *pdfplumber*. Used when tabular grid lines are detected to perfectly extract Indian balance sheets.
- **Scanned Page Fallback:** *Tesseract OCR*. Activated automatically if no text is found in a PDF layer.

Why not use LLM for ingestion? Feeding entire raw PDFs as images (GPT-4 Vision) is prohibitively slow and lacks deterministic table structuring for 200-page Annual Reports.

2. Dual-Track Financial Extraction Engine

Role: Identifies and extracts exactly 30+ financial metrics (Revenue, PAT, Net Worth, EBITDA, etc.) from Annual Reports.

Technical Implementation & Models Used:

Track A: AI Schema Extraction	Track B: Regex Truth Engine
Model: Claude 3.5 Sonnet API	**Model:** Python RegEx via *ai_analyzer.py*
How it works: We feed the cleaned text into Claude with a strict JSON schema prompt. It leverages spatial awareness to find specific entries (e.g., 'Total Equity' only in the Consolidated Balance Sheet).	**How it works:** Sweeps the document checking for strict accounting truths using 10+ explicit functions with fallback logic.
Indian Context: Prompted specifically to understand Lakhs/Crores scaling natively.	**Indian Context:** Detects document scaling (Crores) in headers. Handles bracket negatives typical of Indian accounting: `(22,400)` = -22400.
Why chosen: Claude 3.5 Sonnet beats GPT-4o at strictly adhering to massive JSON object schemas and zero-shot spatial reasoning in unstructured text.	**Why chosen:** 100% deterministic, zero API cost. Algebraic variables like 'Revenue Growth' are strictly calculated mathematically here to prevent LLM hallucination.

3. Cross-Verification & Fraud Detection Engine

Role: Reconciles user-declared data (Annual report / GST) against ground-truth transactional data (Bank Statements) to detect fraud.

Technical Implementation & Algorithms:

- **Circular Trading Detection:** Uses Pandas to cross-reference credit and debit transactions on the same day. If `credit == debit` and amount > ■1 Lakh, it flags potential round-tripping.
- **Revenue Inflation Analysis:** Computes the Coefficient of Variation (CV) on monthly GST revenues. If CV > 0.5, it flags the revenue as highly irregular/lumpy.
- **Reconciliation Score:** A weighted mathematical formula mapping Annual Report Revenue vs GST Revenue vs Bank Credit summations. Variances >10% trigger immediate referral flags.

4. Automated Secondary Research Agent

Role: Replaces manual Google searches. Aggregates external intelligence regarding the company and its promoters.

Technical Implementation:

- **Web Scraping:** Uses the `duckduckgo-search` library combined with `BeautifulSoup` to bypass traditional API limits.
- **Targeted Queries:** The engine autonomously structures deep-search queries: appending 'NCLT', 'litigation', 'wilful defaulter' to the company/promoter names.
- **Regulatory Crawl:** Queries the MCA (Ministry of Corporate Affairs) and RBI circular domains specifically for penalty or strike-off notices.

5. NLP & Sentiment Analysis Engine

Role: Digests the hundreds of web-scraped articles from the Research Agent to determine risk and sentiment.

Technical Implementation & Models Used:

- **Model:** Custom Indian Credit Lexicon + Heuristic Pipeline.
- **How it works:** Instead of sending 50 news articles to an LLM (which is slow, expensive, and unpredictable), we run a deterministic NLP sweep against 100+ heavily weighted Indian credit keywords (e.g., 'SFIO probe', 'CBI raided', 'IBC resolution').
- **Why chosen over FinBERT or LLMs?** Standard sentiment models like BERT will mark 'Bankrupt' as negative, but they miss Indian lending nuance. A custom lexicon ensures 0 inference latency and 100% precision on regulatory flags.

6. Credit Analysis & Ratios Engine

Role: The underwriting core. Computes ratios and standardizes them into the Five Cs of Credit format.

Technical Implementation:

- **NBFC Routing:** Entity classification auto-detects if the company is an NBFC based on keywords. If True, it routes to an entirely different logic tree computing NII, NIM, PPOP (Pre-Provision Operating Profit), and CRAR.
- **The Five Cs Model (scoring_engine.py):** An established banking heuristic. Weights are algorithmic: Character (25%), Capacity (25%), Capital (20%), Collateral (15%), Conditions (15%).

7. ML Risk Prediction Model

Role: Predicts the probability of default based on tabular financial data and outputs explainable charts.

Technical Implementation & Models Used:

Sub-Component	Technology / Approach
Classifier	*XGBoost Classifier*. Trained on 14 synthesized metrics (icr_ratio, debt_equity, nlp_sentiment_score, etc.). We use XGBoost because it seamlessly handles missing variables (NaN) which are common when scraping disparate PDFs.

Sub-Component	Technology / Approach
Training Data	*Synthetic + PDF Extraction*. The model extracts historical features directly from annual reports via `pdf_training_extractor.py` and augments them using synthetic data centered on real credit distributions to prevent overfitting.
Explainability	*SHAP (SHapley Additive exPlanations)*. Credit scoring cannot be a black box due to RBI regulations. SHAP allows us to output the exact reason a loan was rejected (e.g., 'Score lowered by 15% specifically due to DSCR < 1.0').

8. CAM Generation & Decision Engine

Role: Produces the final downloadable Word and PDF document to be uploaded into LOS (Loan Origination Systems).

Technical Implementation & Models Used:

- **Narrative Synthesis:** *GPT-4o*. We employ GPT-4o exclusively at the very end to act as the 'Credit Officer Writer'. It takes the raw JSON scores from XGBoost and the Regex engine, formatting them into executive, professional banking language.
- **Formatting:** *python-docx* and *reportlab* build the heavily formatted output programmatically, embedding tables, SHAP charts, and risk matrices perfectly every time.

9. Enterprise Readiness & Databricks Data Lake

Role: While the live hackathon prototype processes documents locally for zero-latency, the architecture is inherently built for enterprise deployment using an Apache Spark/Databricks core.

Technical Implementation:

- **databricks_connector.py:** A fully functional module natively querying a Databricks SQL Warehouse to read/write from scalable Delta Tables (*financial_statements, gst_filings, bank_statements, litigation_records*).
- **Persistent Audit Trails:** All final credit decisions, scoring matrices, and feature calculations are automatically pushed back into the enterprise Databricks catalog for historical auditing and continuous ML model retraining.
- **Demo 'Local Fallback' Strategy:** During the pitch demo, we deliberately utilize the connector's *Local Fallback* mode. This prevents the ~2 minute cluster warm-up latency inherent to serverless cloud compute, and proves to the judges that our AI algorithms can execute blazingly fast in pure real-time.